Polar Coordinates and Conversions

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Converting points

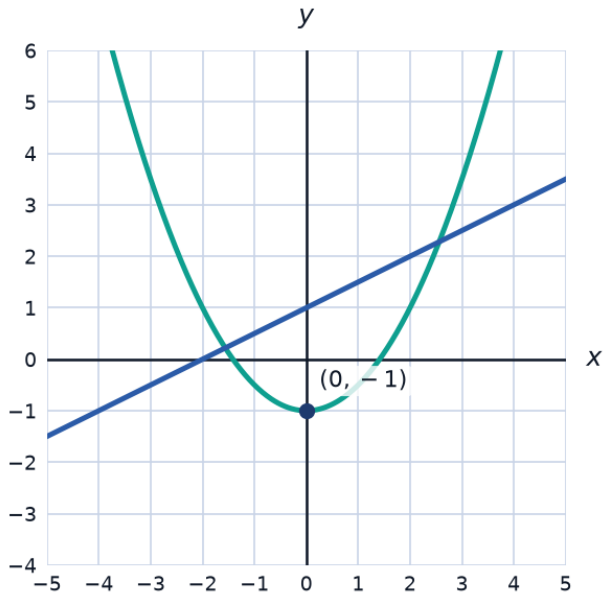
Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$

b $\left(2, \frac{3\pi}{4}\right)$

2 [Foundational] • Target: Converting points

Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:



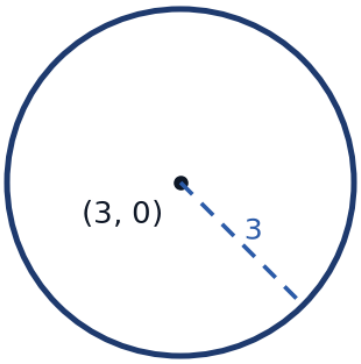
a $(-3, 3)$

b $(0, -5)$

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3 [Foundational] • Target: Converting equations

Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.



4 [Foundational] • Target: Converting equations

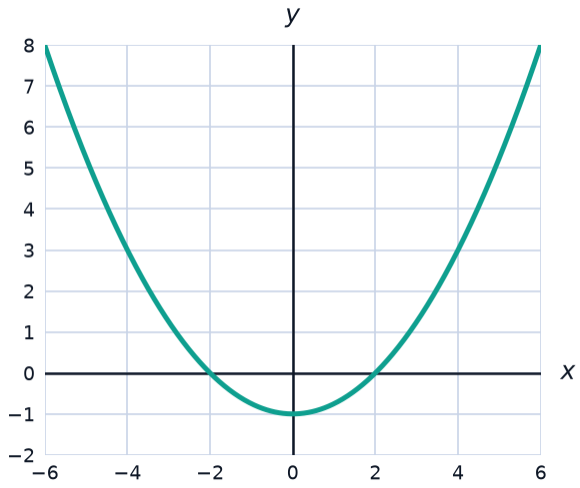
Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$

b $y = x$

5 [Foundational] • Target: Converting equations

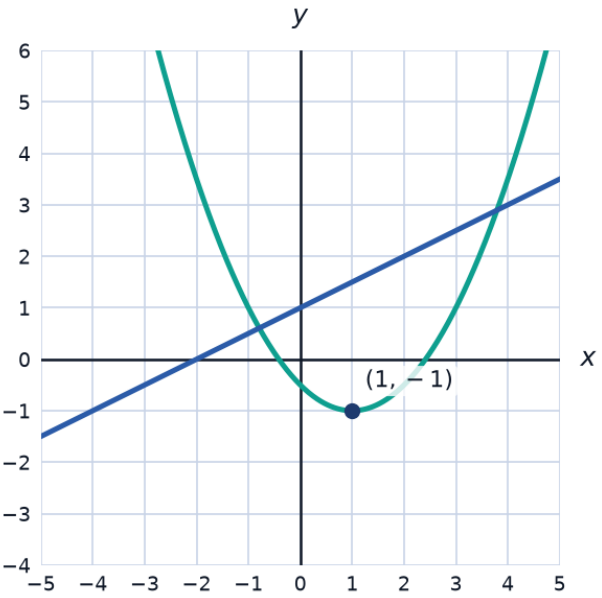
Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.



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6 [Foundational] • Target: Multiple representations

Give two other pairs (r, θ) that represent the same point as $(3, \frac{\pi}{4})$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.



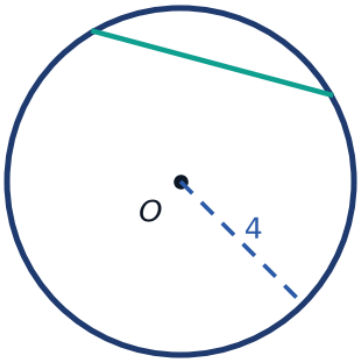
7 [Foundational] • Target: Multiple representations

Plot-reasoning: explain why the polar point $(-2, \frac{\pi}{2})$ corresponds to the same location as $(2, \frac{3\pi}{2})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distance between polar points

Convert $(4, \frac{\pi}{6})$ and $(4, \frac{2\pi}{3})$ to rectangular coordinates, then find the distance between them, to two decimal places.



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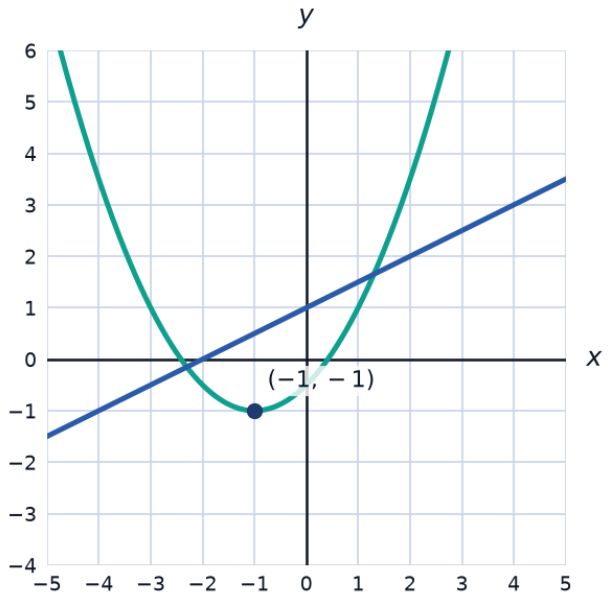
9 [Intermediate] • Target: Further conversion practice

Convert each polar point to rectangular coordinates (exact values):

a $(6, \pi)$ **b** $\left(5, \frac{5\pi}{6}\right)$

10 [Intermediate] • Target: Further conversion practice

Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.



11 [Intermediate] • Target: Why polar coordinates are not unique

Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.

12 [Intermediate] • Target: Why polar coordinates are not unique

The pole (origin) is a special case in polar coordinates. Explain why.

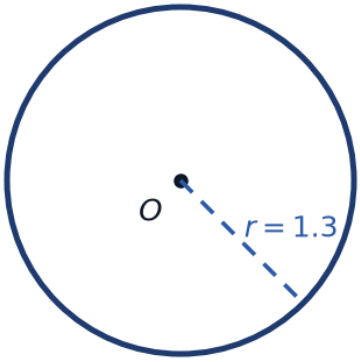
13 [Intermediate] • Target: Converting more complex equations

Convert the rectangular equation $x^2 - y^2 = 4$ (a hyperbola) to polar form.

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14 **[Intermediate]** • **Target: More point conversions with negative r**

Convert the polar point $\left(-4, \frac{\pi}{6}\right)$ (negative r) to rectangular coordinates.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: More point conversions with negative r** • **Cross-Lesson: 2.01 Right-Triangle Trigonometry**

Convert the rectangular point $(-4, -4)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.

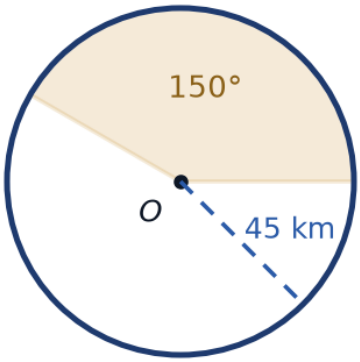
16 **[Challenging]** • **Target: Converting a spiral equation** • **Cross-Lesson: 2.02 Radian Measure and Arc Length**

The Archimedean spiral $r = \theta$ has no simple rectangular equivalent, but explain qualitatively why it spirals outward as θ increases.

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17 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A radar station models detected objects using polar coordinates (r, θ) where r is distance in km and θ is bearing angle. An aircraft is detected at $(r, \theta) = (45, \frac{5\pi}{6})$. Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.



18 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

A second aircraft is detected at rectangular coordinates $(30, -40)$ relative to the same radar station. Convert to polar coordinates (r, θ) with $r > 0$ and $0 \le \theta < 2\pi$, to two decimal places.

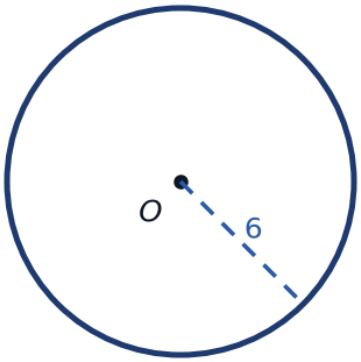
19 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Convert the rectangular equation of a vertical line, $x = 5$, to polar form, and explain why the result is not simply a constant r.

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20 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Convert the polar equation $r = 6$ to rectangular form, and explain in one sentence why this equation contains no θ term.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Plotting with Positive and Negative Radii

How is a point with negative radius $(-r, \theta)$ plotted in polar coordinates?

a. Plot along the opposite ray directly across the pole (angle $\theta + \pi$)

b. Plot in the negative y direction

c. Reflect across the y-axis

d. The point is undefined

22 [Foundational] • Target: Plotting with Positive and Negative Radii

Which polar coordinate represents the same point as $(3, \pi/3)$?

a. $(-3, 4\pi/3)$

b. $(-3, 2\pi/3)$

c. $(3, 4\pi/3)$

d. $(-3, -\pi/3)$

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23 [Foundational] • Target: Plotting with Positive and Negative Radii

In which quadrant does the polar point $(4, 7\pi/6)$ lie?

a. Quadrant III

b. Quadrant II

c. Quadrant IV

d. Quadrant I

24 [Foundational] • Target: Converting Polar to Cartesian Coordinates

State the formulas to convert polar coordinates (r, θ) to Cartesian coordinates (x, y).

a. $x = r\cos\theta, y = r\sin\theta$

b. $x = r\sin\theta, y = r\cos\theta$

c. $x = r^2\cos\theta, y = r^2\sin\theta$

d. $x = \frac{\cos\theta}{r}, y = \frac{\sin\theta}{r}$

25 [Foundational] • Target: Converting Polar to Cartesian Coordinates

Convert the polar point $(6, 2\pi/3)$ to exact Cartesian coordinates.

a. $(-3, 3\sqrt{3})$

b. $(3, -3\sqrt{3})$

c. $(-3\sqrt{3}, 3)$

d. $(-3, -3\sqrt{3})$

26 [Foundational] • Target: Converting Polar to Cartesian Coordinates

Convert $(4, -\pi/4)$ to Cartesian coordinates.

a. $(2\sqrt{2}, -2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(2\sqrt{2}, 2\sqrt{2})$

d. $(4, -4)$

27 [Foundational] • Target: Converting Cartesian to Polar Coordinates

Convert the Cartesian point $(-2, 2)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(2\sqrt{2}, 3\pi/4)$

b. $(2\sqrt{2}, 5\pi/4)$

c. $(4, 3\pi/4)$

d. $(2\sqrt{2}, \pi/4)$

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28 [Foundational] • Target: Converting Cartesian to Polar Coordinates

Convert $(0, -5)$ to polar coordinates.

a. $(5, 3\pi/2)$

b. $(5, \pi/2)$

c. $(-5, 0)$

d. $(5, \pi)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Converting Cartesian to Polar Coordinates

Convert $(3, -3\sqrt{3})$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(6, 5\pi/3)$

b. $(6, 4\pi/3)$

c. $(6, 2\pi/3)$

d. $(12, 5\pi/3)$

30 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

Convert the polar equation $r = 4\cos\theta$ to Cartesian form.

a. $(x-2)^2 + y^2 = 4$

b. $x^2 + y^2 = 4$

c. $x^2 + (y-2)^2 = 4$

d. $x = 4$

31 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

Convert $r = 5\csc\theta$ to Cartesian form.

a. $y = 5$

b. $x = 5$

c. $x^2 + y^2 = 25$

d. $y = 5x$

32 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

Convert the line $y = 3x$ to a polar equation.

a. $\theta = \arctan(3)$

b. $r = 3$

c. $r = 3\cos\theta$

d. $\theta = 3$

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33 **[Intermediate]** • **Target: Converting Cartesian to Polar Coordinates**

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. 5

b. 7

c. 25

d. 16

34 **[Intermediate]** • **Target: Converting Equations Between Polar and Cartesian**

What is the Cartesian equation of the polar curve $r = 10$?

a. $x^2 - y^2 = 100$

b. $x^2 + y^2 = 100$

c. x+y=100

d. $x^2 + y^2 = 10$

35 **[Intermediate]** • **Target: Converting points**

What is the Cartesian equation of the polar curve $r = 11$?

a. $x^2 + y^2 = 121$

b. $x^2 + y^2 = 11$

c. $x^2 - y^2 = 121$

d. x+y=121

36 **[Intermediate]** • **Target: Converting equations**

What is the Cartesian equation of the polar curve $r = 12$?

a. $x^2 + y^2 = 12$

b. $x^2 - y^2 = 144$

c. x+y=144

d. $x^2 + y^2 = 144$

37 **[Intermediate]** • **Target: Multiple representations**

What is the Cartesian equation of the polar curve $r = 13$?

a. $x^2 + y^2 = 13$

b. $x^2 - y^2 = 169$

c. x+y=169

d. $x^2 + y^2 = 169$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Distance between polar points • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

What is the Cartesian equation of the polar curve $r = 14$?

a. $x^2 + y^2 = 196$

b. x+y=196

c. $x^2 - y^2 = 196$

d. $x^2 + y^2 = 14$

39 [Challenging] • Target: Further conversion practice • Cross-Lesson: 3.01 The Unit Circle and Exact Values

What is the Cartesian equation of the polar curve $r = 15$?

a. $x^2 + y^2 = 225$

b. $x^2 + y^2 = 15$

c. x+y=225

d. $x^2 - y^2 = 225$

40 [Challenging] • Target: Why polar coordinates are not unique • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

What is the Cartesian equation of the polar curve $r = 16$?

a. $x^2 + y^2 = 16$

b. x+y=256

c. $x^2 - y^2 = 256$

d. $x^2 + y^2 = 256$

41 [Challenging] • Target: Converting more complex equations • Cross-Lesson: 3.03 Sum and Difference Formulas

What is the Cartesian equation of the polar curve $r = 17$?

a. x+y=289

b. $x^2 + y^2 = 289$

c. $x^2 - y^2 = 289$

d. $x^2 + y^2 = 17$

42 [Challenging] • Target: More point conversions with negative r • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the Cartesian equation of the polar curve $r = 18$?

a. x+y=324

b. $x^2 - y^2 = 324$

c. $x^2 + y^2 = 324$

d. $x^2 + y^2 = 18$

43 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 4.01 Circles and Parabolas

What is the Cartesian equation of the polar curve $r = 19$?

a. x+y=361

b. $x^2 - y^2 = 361$

c. $x^2 + y^2 = 361$

d. $x^2 + y^2 = 19$

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44 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 4.02 Ellipses

What is the Cartesian equation of the polar curve $r = 20$?

a. $x^2 - y^2 = 400$

b. x+y=400

c. $x^2 + y^2 = 400$

d. $x^2 + y^2 = 20$

45 [Challenging] • Target: Plotting with Positive and Negative Radii • Cross-Lesson: 4.03 Hyperbolas

What is the Cartesian equation of the polar curve $r = 4$?

a. $x^2 - y^2 = 16$

b. x+y=16

c. $x^2 + y^2 = 4$

d. $x^2 + y^2 = 16$

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